

Homework 3

1. Introduction

Face de-identification is the process of modifying facial images to prevent automatic recognition while preserving visual utility. This homework tests the security effectiveness of traditional de-identification techniques and demonstrates that AI-based attacks can bypass them, before introducing differential privacy as principled defense.

The assignment has the following three steps:

1. Apply the face de-identification using pixelization and Gaussian blur at multiple parameter settings
2. Train a CNN model to re-identify the subjects from the de-identified images
3. Use differential privacy noise to defend against the CNN attack, and evaluate the privacy-utility trade off.

2. Step 1 : Face De-Identification

2.1. Methods

Pixelization : The image is divided into non overlapping $b \times b$ blocks. Each block is replaced by its mean pixel value. Larger block size b produces stronger de-identification but lower visual quality.

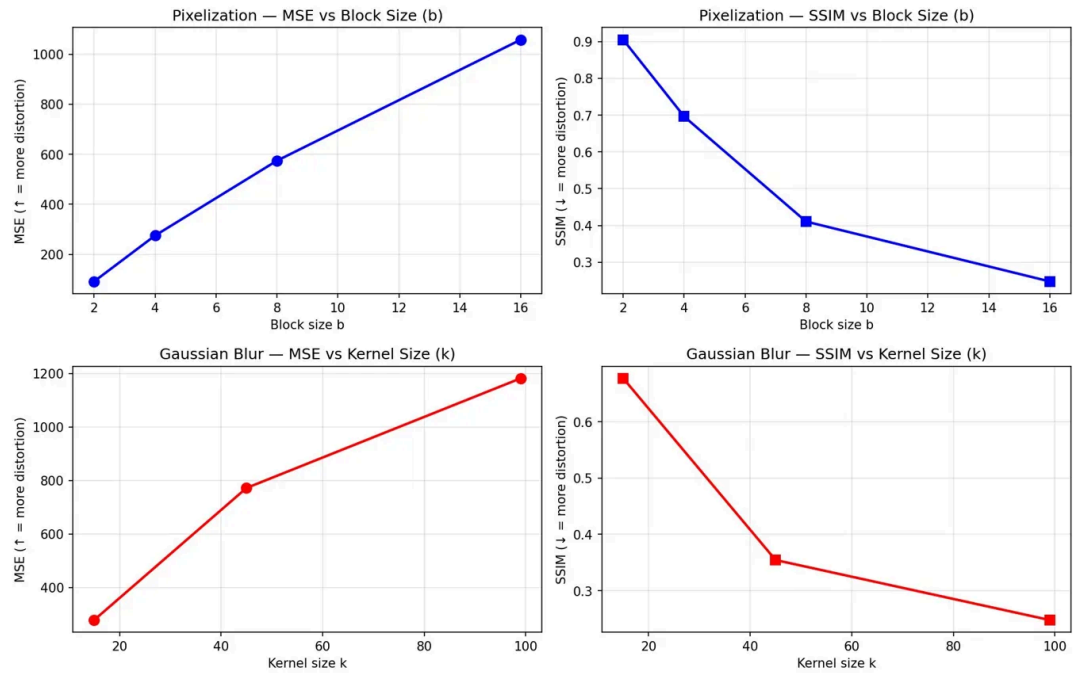
Gaussian Blur : A $k \times k$ Gaussian-weighted filter is applied to the image. For each pixel, the output is the weighted average of its $k \times k$ neighborhood. Larger kernel size produces stronger smoothing and more identity concealment

2.2. Visualization



2.3. Quality Metrics

Step 1: Visual Quality Metrics



3. Step 2 : CNN Re-Identification Attack

3.1. Model Architecture

The attack model is a CNN with three convolutional blocks followed by two connected layers. Each convolutional block consists of a Conv2dLayer, Batch Normalization, ReLU activation, and Max Pooling. The classifier uses Dropout ($p=0.5$) to reduce overfitting.

- Input : 64x64 grayscale image
- Conv Block 1 : 32 filters, 3x3 kernel -> MaxPool -> 32x32
- Conv Block 2 : 64 filters, 3x3 kernel -> MaxPool -> 16x16
- Conv Block 2 : 128 filters, 3x3 kernel -> MaxPool -> 8x8
- FC Layer : 128x8x8 -> 256 -> 40 classes (subjects)

3.2. Training Setup

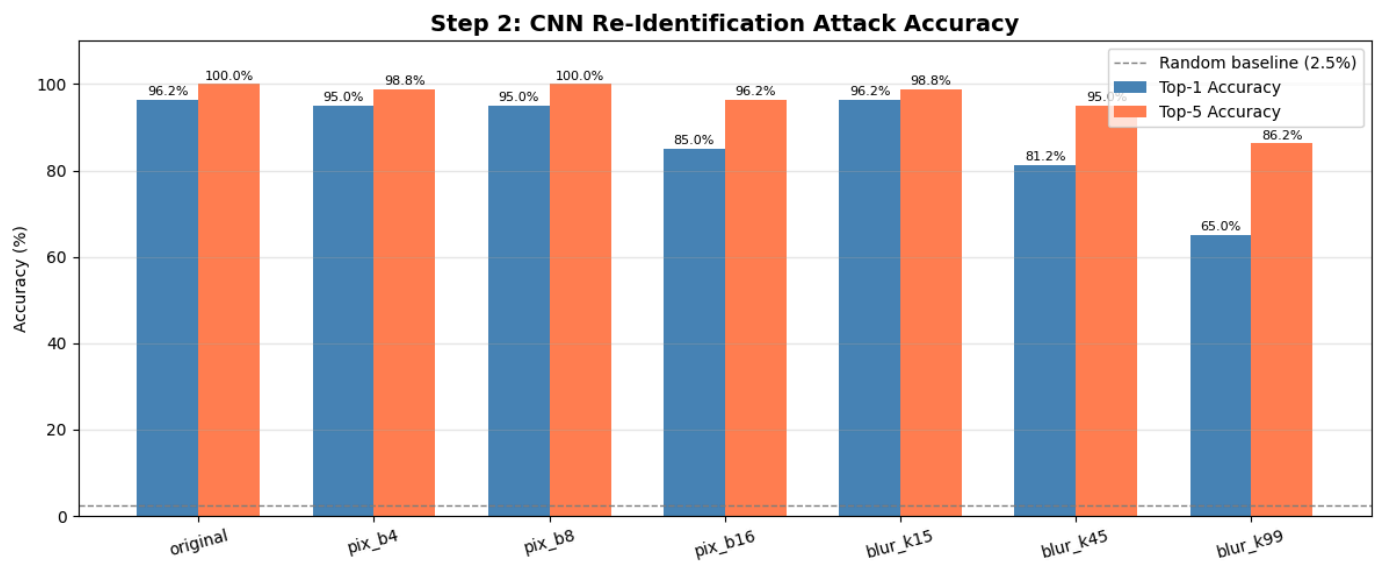
- Dataset split: 8 images per subject for training, 2 for testing (stratified)
- Optimizer: Adam ($lr=0.001$, $weight_decay=1e-4$)
- Loss: Cross-Entropy
- Epochs: 50
- LR Scheduler: StepLR (halve every 20 epochs)
- One model is trained per de-identification variant

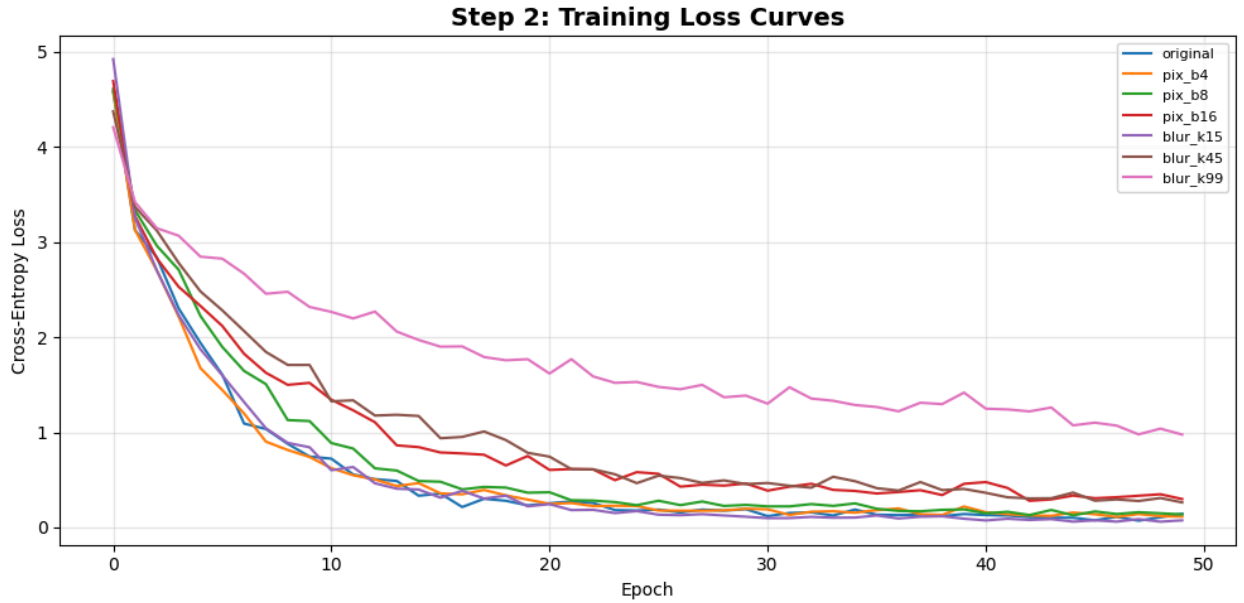
3.3. Results

A separate CNN model is trained and evaluated for each de-identification variant.

The baseline accuracy corresponds to random guessing among 40 subjects.

Dataset	Baseline	Original	Pix b=4	Pix b=8	Pix b=16	Blur k=15	Blur k=45	Blur k=99
Top-1	2.5%	96.25%	95%	95%	85%	96.25%	81.25%	65%
Top-5	12.5%	100%	98.75%	100%	96.25%	98.75%	95%	86.25%





4. Step 3 : Defense via Differential Privacy

4.1. Implementation

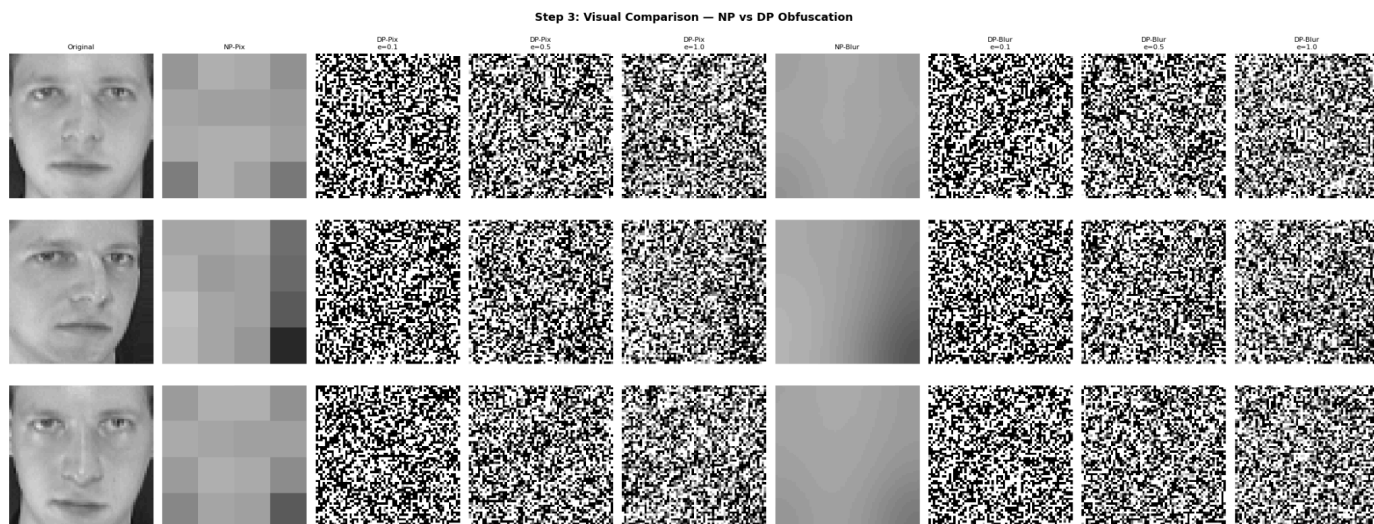
Laplace Mechanism: Laplace noise with scale = sensitivity / epsilon is added to each pixel. The sensitivity is set to 255 (full pixel range). The noisy pixel values are clamped to $[0, 255]$.

Two DP variants are implemented, both using the strongest non-private baseline settings:

- DP-Pix (b=16): Pixelize first (b=16), then add Laplace noise
- DP-Blur (k=99): Gaussian blur first (k=99), then add Laplace noise

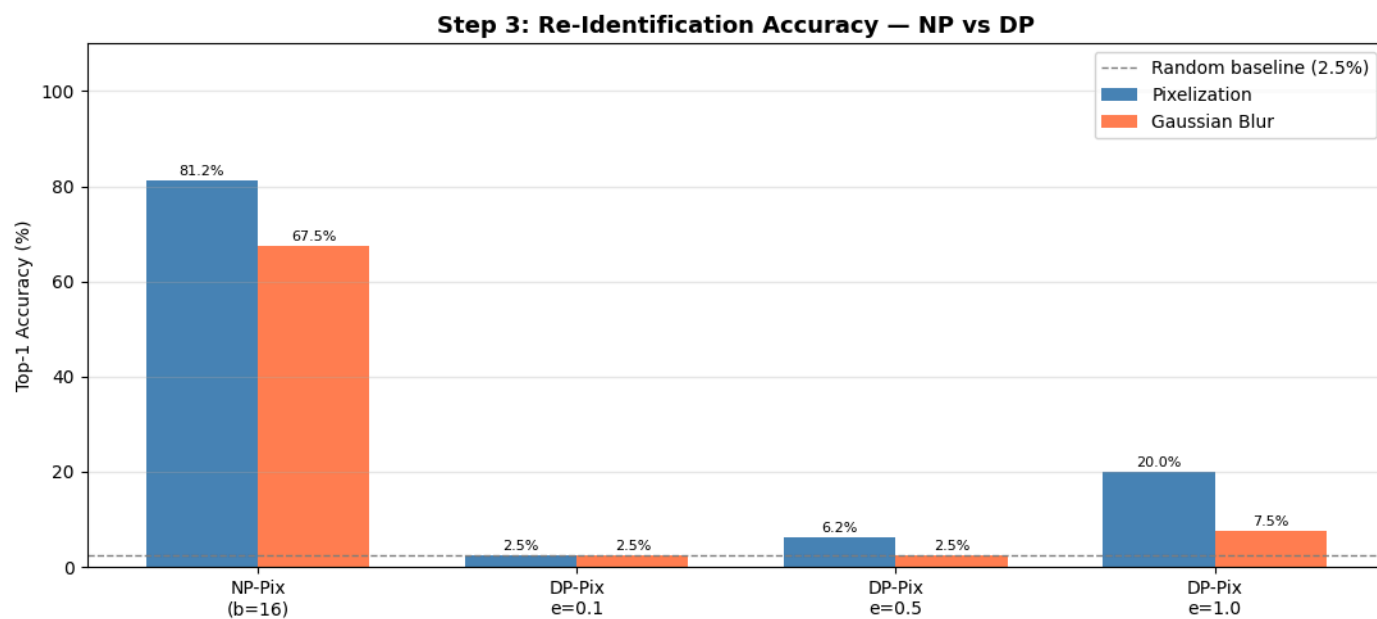
Epsilon values tested: 0.1, 0.5, 1.0 (smaller = more noise = stronger privacy)

4.2. Visualization



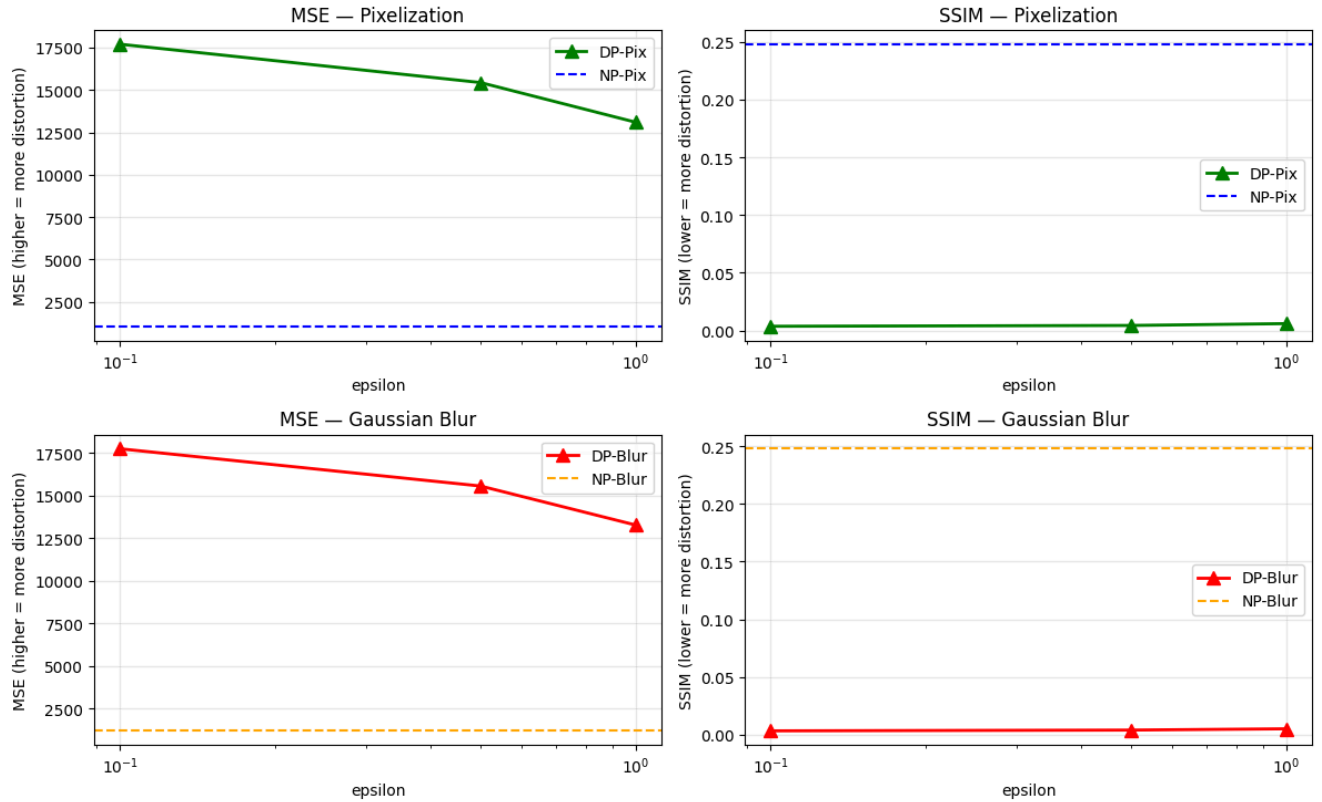
4.3. CNN Attack Results under DP

Dataset	Baseline	NP-Pix b=16	DP-Pix e=0.1	DP-Pix e=0.5	DP-Pix e=1.0	NP-Blur k=99	DP-Blur e=0.1	DP-Blur e=0.5	DP-Blur e=1.0
Top-1	2.5%	81.25%	2.50%	6.25%	20.00%	67.50%	2.50%	2.50%	7.50%
Top-5	12.5%	96.25%	12.50%	16.25%	56.25%	88.75%	12.50%	12.50%	28.75%



4.4. Privacy-Utility trade off

Step 3: Utility vs Privacy Budget (epsilon)
Smaller epsilon = stronger privacy = more distortion



5. Conclusion

This homework demonstrated a complete attack-and-defense pipeline for face de-identification:

- Step 1 showed that both pixelization and Gaussian blur successfully obscure facial identity visually, with stronger parameters causing greater image distortion (higher MSE, lower SSIM).
- Step 2 revealed that a simple CNN can re-identify subjects from de-identified images with accuracy far above the random baseline, proving that traditional de-identification methods alone are insufficient privacy guarantees.
- Step 3 showed that adding differential privacy noise (Laplace mechanism) on top of traditional de-identification significantly reduces CNN attack accuracy, approaching the random baseline at small epsilon values. However, this comes at the cost of image quality, illustrating the fundamental privacy-utility trade-off in differential privacy.